Sanae Faghfouri

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Summary

Motivated third-year Computer Science student currently completing a co-op work term, with strong experience in applied research, backend and full-stack development, and database-driven systems. Proficient in Java, C#, TypeScript, SQL Server, Spring Boot, React, and .NET. Experienced in designing and evaluating software systems, developing and documenting APIs, and supporting data-driven and machine learning–related workflows. Proven ability to collaborate across technical and non-technical teams. Research Director at AI4Impact, with a focus on responsible and impactful AI applications.

Education

York University — B.A. (Honours) Computer Science 2023–Present
Toronto, Canada
Relevant Coursework: Software Design, Advanced Object-Oriented Programming, Software Tools, Database Systems, Fundamentals of Data Structures, Design and Analysis of Algorithms, Computer Organization, Discrete Mathematics for Computer Science, Theory of Computation.

University of Tabriz — B.Sc. Computer Science Sep 2022 – Aug 2023
Tabriz, Iran
Relevant Coursework: General Mathematics I & II, Discrete Mathematics, Computer Programming I & II.

Skills

Programming: Java, C#, TypeScript, JavaScript, Python, C, C++, SQL, HTML, CSS
Frameworks & Libraries: Spring Boot, .NET, React, Vite, Node.js (basics)
Databases: Microsoft SQL Server, Azure SQL
Tools: Git, GitHub, Docker, Swagger, Postman, Azure Data Studio, VS Code, Eclipse
Concepts: REST APIs, MVC architecture, CRUD operations, Data modeling, QA, UAT
Operating Systems: macOS, Windows, Linux

Projects

BookBuddy — Full-Stack Reading Tracker Java, Spring Boot, React, SQL, Docker

- Built backend using layered architecture (Controller, Service, Repository) with Spring Boot.
- Designed SQL Server schema and JPA entity relationships.
- Developed REST endpoints for users, books, progress tracking, and analytics.
- Tested APIs using Swagger UI and Postman.
- Built frontend with React + TypeScript using modular components.

Billy — Expense Sharing & Transaction Management System Java, SQL, JDBC

- Developed a backend system to manage shared expenses, supporting transaction creation, splitting, and tracking between multiple users.
- Implemented transactional operations using JDBC with commit/rollback mechanisms to ensure data consistency and fault tolerance.
- Designed relational database schema for expenses, users, and split allocations.
- Built service-layer logic to handle expense creation and automated cost distribution across participants.
- Ensured data integrity through validation checks and controlled database interactions.

MyPlanner — Productivity & Task Management App TypeScript, JavaScript, HTML/CSS

- Developed a task manager with filtering, editing, sorting, and priority features.
- Implemented persistent storage using the Local Storage API.
- Designed a responsive interface using HTML/CSS and DOM manipulation.

Experience

IT Project Coordinator (Co-op) — HSS Business Solutions Jan 2026–Present

- Coordinating QA, UAT, and release readiness activities across internal systems and enterprise applications.
- Developed a C#/.NET application to support report migration processes in Cognos, improving data consistency and migration efficiency.
- Contributed to both development and QA efforts, working across the full software lifecycle to support IT team deliverables.
- Performed manual and automated testing to validate functionality, data integrity, and system behavior across multiple environments.
- Wrote and maintained test scripts and test cases for regression and UAT workflows using tools such as ACCELQ and BrowserStack.
- Supported business analysis by gathering requirements, clarifying use cases, and defining acceptance criteria.
- Verified backend logic and data accuracy through direct interaction with server-side systems and relational databases.
- Collaborated with developers, QA analysts, and stakeholders to track defects, validate fixes, and ensure high-quality releases.

Research Director — AI4Impact Club, York University Oct 2025–Present

- Leading a research initiative focused on applying foundation models to brain tumor detection using MRI data.
- Conducting literature reviews and identifying gaps in CNN-based and transfer learning approaches.
- Designing comparative benchmarks between task-specific neural networks and multimodal foundation models.
- Defining experimental setups involving multi-dataset evaluation and cross-scanner generalization.
- Evaluating models using metrics such as AUC, accuracy, precision, recall, F1-score, and false-negative rate.

Research Assistant — SEIR Model–Driven Epidemiology (EpiMDE), EASE Lab, York University 2025–Present

Under the supervision of Prof. Marios Fokaefs

- Conducting research on model-driven epidemiological systems using SEIR-based compartmental models.
- Working with metamodels (Ecore) and XML-based representations to formally define disease spread models.
- Designing and validating epidemiological model structures using structured inputs such as tables and graphs.
- Supporting evaluation of AI-generated model outputs by measuring structural correctness, precision, recall, and hallucination rates.
- Collaborating on experimental workflows that integrate software modeling, data-driven inputs, and AI-assisted generation.

Accounting Intern — Roy Parvar Zanjan Company (Part-time) Sep 2022–Apr 2023

- Supported bookkeeping, invoicing, and month-end financial reconciliations using Excel.
- Prepared and reviewed expense reports, tracked financial transactions, and maintained organized records.
- Assisted with accounts payable and accounts receivable tracking, including invoice verification.
- Reconciled receipts and supporting documents to ensure accuracy and completeness of financial data.
- Worked closely with accounting staff to ensure data accuracy, compliance with internal procedures, and timely reporting.

Volunteering

SCLD Volunteer — York University May 2025–Sep 2025
Event Volunteer — Capstone Day 2025, York University
Event Volunteer — Discovery Day 2024, York University